

**XAI Machine Learning-Based Fault
Classification of Power Transformers Using
Dissolved Gas Analysis (DGA)**

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**THIS REPORT IS SUBMITTED IN FULFILMENT
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XAI MACHINE LEARNING-BASED FAULT CLASSIFICATION OF
POWER TRANSFORMERS USING DISSOLVED GAS ANALYSIS
(DGA)

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DECLARATION OF THESIS

Title of thesis

XAI MACHINE LEARNING-BASED FAULT CLASSIFICATION OF
POWER TRANSFORMERS USING DISSOLVED GAS ANALYSIS
(DGA)

I MOHAMAD SUHAIMIE BIN ABDUL GAFAR HUSSIN

I hereby declare that the content of this thesis is my own work, except for quotations, equations, summaries, and references, which have been appropriately acknowledged. Additionally, if Generative Artificial Intelligence has been used, its application and purpose have been transparently disclosed in the disclosure section below

I acknowledge that during the preparation of this thesis, I have used Large Language Models (LLMs) as a supplementary tool to assist in various aspects of the research and writing process. Specifically, LLMs were used to improve language clarity, enhance coherence, and support the structuring of content. All outputs generated were critically reviewed, verified, and aligned with the research objectives and technical requirements of this study.

Thursday 18th June, 2026

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ABSTRACT

Power transformers are critical components in electrical power systems, and their failure can lead to severe operational disruptions and significant economic losses. Dissolved Gas Analysis (DGA) is a widely used diagnostic technique for detecting incipient faults by analysing key gases such as hydrogen (H_2), methane (CH_4), and acetylene (C_2H_2). However, conventional rule-based interpretation methods are limited in their ability to handle complex and overlapping fault conditions.

This study proposes an Explainable Artificial Intelligence (XAI)-based machine learning framework for transformer fault classification using DGA data. The framework integrates advanced machine learning algorithms with systematic data preprocessing and feature engineering to improve classification performance under challenging conditions such as class imbalance and limited labelled data.

To enhance interpretability, SHapley Additive exPlanations (SHAP) is employed to provide feature-level explanations of model predictions, ensuring alignment with known physical fault mechanisms. This enables both global and local understanding of model behaviour.

The proposed framework aims to achieve a balanced trade-off between accuracy, robustness, and interpretability, making it suitable for practical deployment in power system condition monitoring and fault diagnosis applications.

ABSTRAK

Transformer kuasa merupakan komponen kritikal dalam sistem tenaga elektrik, dan kegagalannya boleh menyebabkan gangguan operasi yang serius serta kerugian ekonomi yang besar. Analisis Gas Terlarut (DGA) ialah teknik diagnostik yang digunakan secara meluas untuk mengesan kerosakan awal dengan menganalisis gas seperti hidrogen (H_2), metana (CH_4), dan asetilena (CH_2H_2). Walau bagaimanapun, kaedah berasaskan peraturan tradisional mempunyai keterbatasan dalam menangani keadaan kerosakan yang kompleks dan bertindih.

Kajian ini mencadangkan rangka kerja pembelajaran mesin berasaskan Explainable Artificial Intelligence (XAI) untuk klasifikasi kerosakan transformer menggunakan data DGA. Rangka kerja ini menggabungkan algoritma pembelajaran mesin dengan pemprosesan data dan kejuruteraan ciri bagi meningkatkan prestasi klasifikasi dalam keadaan cabaran seperti ketidakseimbangan kelas dan data berlabel yang terhad.

Bagi meningkatkan kebolehjelasan, SHapley Additive exPlanations (SHAP) digunakan untuk memberikan penjelasan pada tahap ciri terhadap ramalan model, sekali gus memastikan keselarasan dengan mekanisme fizikal kerosakan. Ini membolehkan pemahaman global dan tempatan terhadap tingkah laku model.

Rangka kerja yang dicadangkan bertujuan mencapai keseimbangan antara ketepatan, kebolehpercayaan, dan kebolehjelasan, menjadikannya sesuai untuk aplikasi sebenar dalam pemantauan keadaan dan diagnostik sistem kuasa.

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AVT Auditory Vigilance Task

List of Symbols

CHAPTER 1

INTRODUCTION

1.1 Overview of the Chapter

This chapter introduces the research background and foundational concepts related to explainable machine learning–based fault classification of power transformers using Dissolved Gas Analysis (DGA). The purpose of this chapter is to establish the context of the study, define the problem domain, and highlight the motivation for integrating machine learning and explainable artificial intelligence in transformer fault diagnosis.

Power transformers are essential components in modern electrical power systems, and their continuous operation is critical for ensuring reliable energy transmission and distribution. Due to their importance, even minor faults can lead to significant system instability and operational losses. Therefore, effective condition monitoring techniques are required to ensure early fault detection and system reliability (Reference details not provided, n.d.-k).

This chapter also provides a structured transition from traditional diagnostic approaches to data-driven machine learning methods. It explains how conventional DGA interpretation techniques are limited in handling complex fault patterns, and how machine learning models offer improved adaptability through data-driven learning mechanisms (Reference details not provided, n.d.-d).

Finally, the chapter outlines the key research gaps, objectives, and scope of the study, which collectively guide the development of the proposed methodology in subsequent chapters (Reference details not provided, n.d.-c).

1.2 Background of Transformer Condition Monitoring

Power transformers operate under continuous electrical, thermal, and mechanical stress throughout their operational lifespan. These stresses gradually degrade insulation materials, making transformers susceptible to internal faults if not properly monitored.

Transformer failures can lead to severe consequences such as power outages, equipment damage, and cascading failures across interconnected power networks. These

failures may also impact critical infrastructure including healthcare systems, transportation networks, and industrial operations.

To mitigate such risks, condition monitoring systems are widely implemented in power systems. These systems aim to detect early signs of deterioration before catastrophic failure occurs. Among various diagnostic methods, Dissolved Gas Analysis (DGA) has been widely recognized as one of the most effective techniques for transformer health assessment (Reference details not provided, n.d.-a).

1.3 Background of the Problem

Dissolved Gas Analysis (DGA) is a diagnostic technique used to identify internal transformer faults by analyzing gases dissolved in transformer oil. When transformers experience thermal or electrical stress, decomposition of insulating materials produces characteristic gases that dissolve in the oil.

The key gases involved in DGA include hydrogen (H_2), methane (CH_4), ethane (C_2H_6), ethylene (C_2H_4), acetylene (C_2H_2), carbon monoxide (CO), and carbon dioxide (CO_2). These gases are directly associated with different fault mechanisms and energy levels inside transformers (International Electrotechnical Commission, 2022).

International standards such as IEEE C57.104-2019 and IEC 60599:2022 define structured diagnostic guidelines for interpreting gas concentrations and ratios. These standards are widely used in industrial applications for transformer fault diagnosis (IEEE Power Engineering Society, 2019).

However, real-world transformer environments introduce complexity due to varying load conditions, aging effects, and environmental influences, which may affect gas generation patterns and reduce the effectiveness of rule-based interpretation methods.

1.3.1 Physical Meaning of Gas Generation

Gas generation in transformer oil is directly linked to the type and severity of internal faults. Different fault mechanisms produce distinct gas signatures depending on the energy level of the fault.

Low-energy electrical faults such as partial discharge typically generate hydrogen as the dominant gas due to localized ionization processes. Medium and high thermal faults

58 generate hydrocarbon gases such as methane, ethane, and ethylene due to gradual
59 decomposition of insulation materials.

60 High-energy electrical faults such as arcing generate acetylene, which is considered a
61 critical indicator of severe internal transformer failure (Reference details not provided,
62 n.d.-d).

63 Thermal faults are generally associated with gradual overheating processes, whereas
64 electrical faults are associated with sudden high-energy discharge events. This physi-
65 cal relationship between gas generation and fault types forms the basis of DGA-based
66 transformer diagnosis (Reference details not provided, n.d.-c).

67 **1.4 1.4 Conventional DGA Interpretation Methods**

68 Conventional DGA interpretation methods are rule-based diagnostic approaches used
69 to classify transformer faults based on gas concentrations and ratios. These methods
70 are widely used due to their simplicity and alignment with international standards.

71 Common methods include the Key Gas method, Rogers Ratio method, IEC Ratio method,
72 Duval Triangle, and Duval Pentagon. These methods rely on predefined rules or graph-
73 ical boundaries to classify transformer fault conditions (Reference details not provided,
74 n.d.-a).

75 The main advantage of these methods is their ease of use and interpretability, making
76 them suitable for field applications where quick decision-making is required.

77 However, these methods have limitations such as rigid classification boundaries, sen-
78 sitivity to measurement errors, and inability to handle overlapping or complex fault
79 conditions. In some cases, different methods may produce inconsistent results for the
80 same transformer condition.

81 **1.5 1.5 Machine Learning Motivation in DGA Fault Diagnosis**

82 Machine learning techniques have been introduced to overcome the limitations of con-
83 ventional DGA interpretation methods. These techniques enable data-driven learning
84 of complex nonlinear relationships between gas concentrations and transformer fault
85 types.

86 Unlike rule-based systems, machine learning models can automatically learn patterns
87 from historical data without requiring predefined diagnostic rules. This allows them to

88 adapt more effectively to real-world transformer operating conditions (Reference details
89 not provided, n.d.-i).

90 One major advantage of machine learning is automation, which enables consistent and
91 scalable fault classification across large datasets and monitoring systems.

92 However, challenges such as class imbalance, weak feature representation, and lack of
93 interpretability remain significant issues in DGA-based machine learning systems (Ref-
94 erence details not provided, n.d.-b).

95 **1.5.1 1.5.1 Problem Formulation**

96 Despite the success of machine learning in transformer fault classification, several chal-
97 lenges remain unresolved.

98 Machine learning models often show inconsistent performance across different DGA
99 datasets due to variations in data distribution and preprocessing methods. This reduces
100 model reliability and generalization capability.

101 In addition, DGA datasets frequently suffer from class imbalance and weak feature
102 representation, which negatively affects model learning performance for minority fault
103 classes.

104 Furthermore, many machine learning models lack interpretability, making it difficult to
105 understand how gas features influence final predictions (Reference details not provided,
106 n.d.-h).

107 These challenges highlight the need for an integrated framework that combines feature
108 optimization, consistent model evaluation, and explainable artificial intelligence tech-
109 niques (Reference details not provided, n.d.-e).

110 **1.5.2 1.5.2 Research Gap**

111 The first research gap is the inconsistency in feature representation across studies,
112 where different approaches use raw gas values, gas ratios, or hybrid features without
113 standardized evaluation.

114 The second research gap is the lack of consistent benchmarking across machine learning
115 models, making performance comparison unreliable across studies (Reference details
116 not provided, n.d.-d).

117 The third research gap is the limited integration of explainable artificial intelligence
118 techniques in DGA-based transformer fault classification, reducing interpretability in
119 engineering applications (Reference details not provided, n.d.-I).

120 **1.6 1.6 Hypotheses**

121 This study is guided by the following hypotheses:

- 122 • Different DGA feature sets significantly influence machine learning classification
123 performance.
- 124 • Traditional machine learning models show varying performance under identical
125 datasets.
- 126 • Explainable AI techniques can identify the most influential gas features contribut-
127 ing to transformer fault classification.

128 **1.7 1.7 Research Objectives**

129 The objectives of this study are:

- 130 • To evaluate machine learning models for DGA-based transformer fault classifica-
131 tion.
- 132 • To analyze the impact of different feature sets on classification performance.
- 133 • To apply explainable artificial intelligence techniques to interpret model predic-
134 tions.

135 **1.8 1.8 Scope of Work**

136 This study focuses on DGA-based transformer fault classification using benchmark
137 datasets containing dissolved gas measurements.

138 The gases considered include hydrogen, methane, ethane, ethylene, acetylene, carbon
139 monoxide, and carbon dioxide.

140 The study evaluates traditional machine learning models including SVM, Decision Tree,
141 Random Forest, KNN, and boosting-based methods.

142 Deep learning approaches, real-time deployment systems, and hardware-level trans-
143 former monitoring are excluded from this study.

144 **1.9 1.9 Organisation of Thesis**

145 This thesis is organised into five chapters. Chapter 1 introduces the research back-
146 ground, problem formulation, and objectives. Chapter 2 presents a comprehensive lit-
147 erature review on DGA, machine learning, and explainable artificial intelligence. Chapter
148 3 describes the methodology including dataset preparation, feature engineering, model
149 development, and evaluation methods. Chapter 4 presents results and discussion, while
150 Chapter 5 concludes the study and provides future recommendations.

CHAPTER 2

LITERATURE REVIEW

This chapter reviews existing literature on dissolved gas analysis (DGA) for transformer fault classification using machine learning (ML) and explainable AI (XAI). Section 2.1 introduces DGA and key gases (H_2 , CH_4 , C_2H_6 , C_2H_4 , C_2H_2 , CO , CO_2) used to diagnose incipient faults. Section 2.2 surveys traditional DGA interpretation methods (Key Gas, Dornenburg, Rogers, IEC ratios, Duval triangle/pentagon), noting their reliance on fixed thresholds/ratios and known limitations (conflicting diagnoses, inability to handle low gas levels). Section 2.3 overviews ML's role: ML models can learn complex, nonlinear gas–fault relationships and improve accuracy over conventional rules, but they face challenges like limited data, class imbalance, and interpretability issues. Section 2.4 covers DGA feature representations: raw gas concentrations, conventional gas ratios, and combined feature sets (e.g. log-transformed or normalized values). It also discusses feature-selection approaches (filter/wrapper/embedded, RF importance, SHAP, PCA, genetic algorithms, mutual information, RFE) to identify informative DGA features. Section 2.5 introduces each traditional ML classifier (SVM, Decision Tree, Random Forest, k-NN, Logistic Regression, Gradient Boosting, XGBoost) with definitions, DGA applications (many studies show RF and XGBoost excel), strengths, and limitations. Section 2.6 examines data-related issues (small sample sizes, severe class imbalance, missing gas readings, inconsistent labeling) and appropriate evaluation metrics (accuracy, precision, recall, F1-score, macro-F1, confusion matrix, ROC/AUC). Section 2.7 describes XAI for transformer DGA, focusing on SHAP/TreeSHAP methods. Recent DGA studies applying SHAP are summarized, for example Tochukwu *et al.* (2026) found that SHAP explanations confirmed that the ML model's behavior aligned with physical gas–fault relations, but early (moderate) faults remained hard to distinguish. Table 2.1–2.4 summarize conventional methods, feature types, ML algorithm performance comparisons, and XAI studies. The review identifies a research gap in systematically optimizing feature subsets per algorithm, rigorously comparing classifiers on common DGA datasets, and using XAI (SHAP) to explain the best model's decisions in transformer fault diagnosis.

2.1 2.1 Dissolved Gas Analysis (DGA) for Transformers

DGA is a standard tool for monitoring transformer health by measuring key gas concentrations dissolved in insulation oil. Under fault conditions (overheating or arcing), the oil and cellulose insulation decompose, generating combustible gases. The primary fault gases are hydrogen (H_2), methane (CH_4), ethane (C_2H_6), ethylene (C_2H_4), acetylene (C_2H_2), and the byproducts carbon monoxide (CO) and carbon dioxide (CO_2). IEEE C57.104-2019 and IEC 60599:2022 codify how these gas levels relate to fault types (thermal, electrical, partial discharge) and normal aging, but interpretation requires human expertise or supplementary methods. DGA thus provides early warning of developing faults; as Mirowski and LeCun (2012) note, DGA “is an investigative tool to monitor [transformer] health and to detect impending failures by recognizing anomalous patterns of DGA concentrations”. Because gas generation also depends on transformer size, age, load history and fault duration, no single universal rule applies.

2.2 2.2 Conventional DGA Interpretation Methods

Traditional DGA methods use thresholds or gas ratios to infer fault category. The **Key Gas** method simply identifies the single gas with the highest concentration as the “fault gas” (e.g. an unusually high acetylene implies arcing). The **Dornenburg** and **Rogers** ratio methods calculate fixed gas ratios (e.g. CH_4/H_2 , C_2H_2/CH_4 , C_2H_4/C_2H_6 , etc.) to map to fault regions. The **IEC ratio method** (International Electrotechnical Commission, 2022) uses a subset of the Rogers ratios (omitting C_2H_6/CH_4) to distinguish low- vs. high-temperature thermal faults. **Duval’s triangle** uses the three hydrocarbon gases CH_4 , C_2H_4 , C_2H_2 on a triangular plot to classify faults, while **Duval’s pentagon** extends this by including H_2 and C_2H_6 to better differentiate fault severity (e.g. high-energy vs. low-energy thermal faults). All these methods are simple and provide fault-level explanations. However, they have notable limitations. Ratio methods can give conflicting or missing diagnoses if gas concentrations fall outside expected ranges. For example, Rogers’s method can fail when a ratio’s denominator is near zero, and the triangle method lacks a “normal aging” zone, making threshold setting ad hoc. When multiple methods are applied, as noted by Liu *et al.* (2018) and others, inconsistent or ambiguous fault assignments are common. In summary, conventional methods rely on expert-chosen gas thresholds and often misclassify borderline or multi-fault cases

211 (Table 2.1).

212 **Table 2.1. Conventional DGA interpretation methods.**

Method	Input Gases / Ratios	Fault Types Detected	Strengths	Limitations
Key Gas	Single highest-concentration gas (H ₂ , CH ₄ , C ₂ H ₆ , C ₂ H ₄ , C ₂ H ₂ , CO, CO ₂)	Predominant fault gas type (thermal vs. electrical)	Always yields a classification (simple)	High misdiagnosis rate ($\approx$ 30–50%); ignores mixed faults
Dornenburg Ratio	CH ₄ /C ₂ H ₄ , CH ₄ /C ₂ H ₂ , C ₂ H ₂ /C ₂ H ₄	Broad thermal vs. discharge	Accounts for gas ratios	Over-simplified; uses fixed thresholds
Rogers Ratio	CH ₄ /H ₂ , C ₂ H ₆ /CH ₄ , C ₂ H ₄ /C ₂ H ₆ , C ₂ H ₂ /C ₂ H ₄	Detailed thermal (low/high) and PD	More fault categories than Key Gas	Inconsistent when gases out-of-range; ignores aging
IEC Ratio (60599)	CH ₄ /H ₂ , C ₂ H ₂ /CH ₄ , C ₂ H ₂ /C ₂ H ₄	Low- vs. high-energy faults, PD	Standardized in IEC [IEC 60599:2022]	Cannot subtype within categories; excludes C ₂ H ₆ /CH ₄
Duval Triangle	CH ₄ , C ₂ H ₄ , C ₂ H ₂ (triangular plot)	Six fault zones (PD, low & high thermal/electrical)	Graphical; intuitive visualization	No “normal” region; requires user-set gas limits
Duval Pentagon (1,2)	H ₂ , CH ₄ , C ₂ H ₆ , C ₂ H ₄ , C ₂ H ₂ (five-point plot)	Refines thermal categories (T1, T2, T3)	Extends triangle method (IEEE Power Engineering Society, 2019)	Complex; still manual; may be ignored if gases low

213 *Sources:* Conventional DGA methods and their gas inputs are detailed in standards
 214 and reviews. Limitations (e.g. no normal zone, conflicting outputs) are noted in recent
 215 analyses.

216 **2.3 2.3 Machine Learning in DGA Fault Classification**

217 Machine learning has been widely applied to improve the reliability and automation of
218 DGA interpretation. Unlike fixed-rule methods, ML algorithms can learn complex, non-
219 linear mappings from gas patterns to fault classes. Mirowski and LeCun (2012) showed
220 that nonlinear models such as kernel SVMs or neural networks marginally outperform
221 linear classifiers on DGA data. In practice, many studies now train supervised classifiers
222 on labeled transformer DGA records (healthy vs. various faults). The goal is to raise
223 overall accuracy and detect mixed or incipient faults that rule-based methods miss. In-
224 deed, Dladla and Thango (2024) report that ML algorithms have been used to “improve
225 the accuracy and reliability of conventional methods” for DGA-based fault detection.
226 For example, Ekojono *et al.* (2022) found that Random Forest models significantly
227 outperformed other classifiers and graphical methods on several Duval-triangle-based
228 datasets.

229 However, ML approaches face their own challenges. Transformer DGA datasets tend to
230 be small and highly imbalanced (few severe faults, many normals), which complicates
231 training and evaluation. Gas measurements may be missing or noisy, and labels (fault
232 diagnosis codes) can vary by data source. Most ML models (especially ensembles and
233 deep nets) also lack inherent interpretability: as Lundberg and Lee (2017) observe,
234 high-accuracy models are often complex and “even experts struggle to interpret” their
235 decisions. Thus, there is a need for explainable techniques to justify ML predictions
236 (see Section 2.9).

237 In summary, ML classifiers (e.g. SVM, RF, XGBoost) have demonstrated higher accuracy
238 than traditional DGA rules, but issues like limited data, class skew, and interpretability
239 remain open problems. This motivates research objectives to systematically compare
240 algorithms and apply XAI to the best model.

241 **2.4 2.4 DGA Feature Representation**

242 **2.4.1 2.4.1 Gas Concentration Features**

243 The simplest feature set for ML models is the raw gas concentrations. As noted, a
244 complete DGA sample includes seven key gas features: H_2 , CH_4 , C_2H_6 , C_2H_4 , C_2H_2 , CO ,
245 and CO_2 . (Other gases such as oxygen or nitrogen may be measured but usually remain
246 constant and are excluded from diagnostics.) Using raw concentrations allows the model

247 to directly learn from the actual measurements. This approach has the advantage of
248 preserving all information, including absolute levels and relative magnitudes. However,
249 raw concentrations often have highly skewed, long-tailed distributions (a few samples
250 with very large gas amounts). Mirowski and LeCun (2012) therefore applied a log
251 transform to stabilize this variability. Raw features also vary with transformer rating
252 and age, which may require normalization or inclusion of meta-features (nameplate
253 power, age) to improve modeling.

254 **2.4.2 Gas Ratio Features**

255 Many traditional DGA patterns are defined by ratios of two gases, as in the Rogers
256 and IEC methods. Common ratio features include CH_4/H_2 , $\text{C}_2\text{H}_6/\text{CH}_4$, $\text{C}_2\text{H}_4/\text{C}_2\text{H}_6$,
257 $\text{C}_2\text{H}_2/\text{C}_2\text{H}_4$, as well as $\text{C}_2\text{H}_2/\text{H}_2$ and CO_2/CO (the latter helps separate oil vs. paper
258 degradation). Using ratios has the benefit of highlighting relative gas patterns
259 independent of scale. For example, Rogers ratios effectively classify fault type across
260 transformers of different sizes. In ML models, explicitly including ratios can accelerate
261 learning of these relationships. However, ratio features can be unstable if the de-
262 nominator gas is near zero or below detection limits. In such cases, the ratio may be
263 undefined or noisy. Models must handle such issues (e.g. by imputation or clipping).
264 Overall, gas ratios provide interpretable features linked to domain knowledge, but they
265 may discard some information (absolute gas level) and can generate extreme values.

266 **2.4.3 Combined and Transformed Features**

267 Most ML studies use combinations of raw and derived features. In addition to raw and
268 pairwise ratios, researchers often include composite features such as *total dissolved*
269 *combustible gas* (TDCG) or percentage contributions of each gas to TDCG. Mirowski
270 and LeCun (2012) showed that a logarithmic transformation of raw gas values implicitly
271 encodes ratios (since $\log(A/B) = \log A - \log B$). Thus, log-transformed raw features allow
272 the model to learn ratio-like distinctions. Some works also normalize gas concentrations
273 by transformer size or operating conditions. Feature sets may also include non-gas
274 attributes (transformer age, load, etc.) if available. Given the many possible features,
275 careful selection is critical (see Section 2.4.4). Hybrid feature sets (raw + ratios +
276 domain features) can improve accuracy, but increase dimensionality. Our first objective

277 will examine which feature subset (e.g. raw only, ratios only, or combined) is optimal
278 for each algorithm in DGA classification.

279 **2.4.4 2.4.4 Feature Selection Methods**

280 To reduce dimensionality and improve generalization, various feature-selection tech-
281 niques are applied to DGA data. **Filter methods** (e.g. correlation thresholds, mu-
282 tual information) quickly remove irrelevant features without involving the learning algo-
283 rithm. **Wrapper methods** (e.g. recursive feature elimination) iteratively select subsets
284 by measuring classifier performance. **Embedded methods** (e.g. LASSO, tree-based
285 importance) select features as part of model training. For example, Random Forest
286 provides an importance score for each gas feature, which can rank features by predic-
287 tive value. **SHAP-based selection** uses SHapley values from an ensemble to identify
288 the most influential features on model output. **Principal Component Analysis (PCA)**
289 can combine gas features into orthogonal components, though it sacrifices interpretabil-
290 ity. **Genetic algorithms (GA)** and other search techniques have also been used to
291 find high-performing feature subsets in DGA classification studies. The choice of fea-
292 ture selection interacts with the algorithm: some (like tree ensembles) perform implicit
293 selection, while others (e.g. SVM) may benefit from pre-filtering. Overall, a variety of fil-
294 ter, wrapper, and embedded methods have been proposed to identify the best gas/ratio
295 features. In this work, we will explore multiple selection methods (e.g. RF importance,
296 SHAP values, PCA, GA, MI, RFE) to find the most informative DGA features for each
297 classifier.

298 **2.5 2.5 Machine Learning Classification Algorithms**

299 **2.5.1 2.5.1 Support Vector Machine (SVM)**

300 **Definition:** SVM is a supervised classifier that finds a separating hyperplane in feature
301 space, possibly using kernel functions for nonlinearity. In classification it tries to maxi-
302 mize the margin between classes.

303 **Applications in DGA:** SVM has been used extensively for transformer fault diagnosis
304 by treating DGA samples as labeled vectors. Bacha *et al.* (2012) and others demon-
305 strated SVM's capability to improve on classic DGA interpretation. In the reviewed stud-
306 ies, SVM often appears as one of the most common algorithms (e.g. ~32% of works

in DGA ML surveys) and typically provides robust performance on gas data. Nonlinear kernels (e.g. RBF) allow SVM to capture complex gas–fault patterns.

Strengths: SVM can handle high-dimensional features and is effective even with limited samples. Kernels make it flexible for complex boundaries. It is less prone to overfitting when parameters are tuned, and it provides a unique optimal solution. SVM also handles non-numeric labels well through one-vs-rest schemes.

Limitations: SVM requires careful choice of kernel and hyperparameters, which can be computationally costly. It does not naturally provide probability estimates (unless methods like Platt scaling are added). On very large datasets (e.g. thousands of samples), SVM training can be slower than tree-based methods. Furthermore, SVM model outputs are not easily interpretable to experts without additional analysis (e.g. feature contributions are obscured).

Linking: In summary, SVM is a strong baseline for DGA classification with proven utility, but its complexity and parameter sensitivity motivate comparing it with more interpretable or automatically tuned algorithms.

2.5.2 Decision Tree (DT)

Definition: A decision tree classifier partitions the feature space by asking a sequence of threshold questions (e.g. “CH₄ > 10 ppm?”), forming a tree of nodes. Leaves correspond to class labels.

DGA Applications: Decision trees have been applied to DGA fault datasets, often in combination with ensemble methods. Their rule-based nature makes them directly analogous to traditional DGA rules (ratio thresholds), but automatically learned from data. For instance, an internal node might split on CH₄/H₂ ratio to decide a thermal vs. arcing fault.

Strengths: Decision trees are simple to interpret: one can trace a path from root to leaf as a human-readable rule (e.g. “IF CH₄/H₂ > 0.2 AND C₂H₂ > 5 ppm THEN fault=T2”). They handle both numerical and categorical data, and can capture nonlinear feature interactions. Training is fast, and they can handle missing values (by surrogate splits).

Limitations: A single DT is prone to overfitting (high variance) – small changes in data can produce different trees. Complex trees with many splits become hard to interpret. In DGA context, a tree might over-specialize to the training set’s gas distributions. Moreover, pure decision trees often underperform ensemble methods in accuracy.

339 **Linking:** Decision Trees serve as a foundation (and as weak learners in boosting).
340 Their interpretable structure is attractive for fault diagnosis, but to achieve high accu-
341 racy on noisy or imbalanced DGA data, ensemble methods (bagging/boosting) or other
342 algorithms are usually preferred.

343 **2.5.3 2.5.3 Random Forest (RF)**

344 **Definition:** Random Forest is an ensemble of decision trees, each trained on a boot-
345 strap sample of the data with random feature selection. Predictions are made by ma-
346 jority voting.

347 **DGA Applications:** RF is one of the most successful algorithms in DGA fault clas-
348 sification. Many studies report RF as achieving the highest accuracy. For example,
349 Ekojono *et al.* (2022) found that RF outperformed SVM, DT, and others on transformer
350 gas datasets. RF handles high-dimensional gas/ratton features well and implicitly ranks
351 feature importance (useful for 2.4.4).

352 **Strengths:** RF significantly reduces overfitting compared to a single tree. It is robust
353 to noise and outliers, and naturally handles missing values. RF provides a built-in esti-
354 mate of feature importance (e.g. which gas features are most predictive). It generally
355 yields high accuracy out-of-the-box without extensive tuning.

356 **Limitations:** The ensemble of many trees is less interpretable than a single tree
357 (though global feature importance can be extracted). RF can be slower to predict on
358 new data (many trees) and requires more memory. It may still struggle if classes are
359 extremely imbalanced (RF tends to favor majority class unless class weights or balanced
360 sampling are used).

361 **Linking:** Because of its strong performance and ease of use, RF is a leading choice
362 for DGA classification. In practice, we will use RF both as a classifier and as a feature
363 selector (via importance scores) in feature optimization.

364 **2.5.4 2.5.4 k-Nearest Neighbors (k-NN)**

365 **Definition:** k-NN is an instance-based classifier: to classify a new sample, it finds the
366 k closest points (in feature space, e.g. Euclidean distance) from the training set and
367 predicts the majority class among them.

368 **DGA Applications:** k-NN has been used for transformer fault detection (sometimes

combined with imputation for missing gas values). Because it is nonparametric, k-NN can model arbitrary decision boundaries by simply storing all examples.

Strengths: k-NN makes no assumptions about the underlying distribution. It automatically adjusts to complex multi-class problems by proximity. It can handle multiclass naturally, and the implementation is conceptually simple.

Limitations: k-NN's performance suffers in high-dimensional spaces due to the "curse of dimensionality" (distance metrics become less meaningful). It is also very sensitive to irrelevant or noisy features unless careful feature selection is done. k-NN requires storing all training data, making prediction slow for large datasets (each query searches through all points). Moreover, because DGA features often vary in scale, k-NN needs proper normalization. Empirical studies often find that k-NN underperforms ensemble methods on DGA; for instance, one study reported k-NN accuracy $\approx 85.7\%$ while RF achieved $\approx 89\%$ on the same data.

Linking: k-NN provides a useful baseline but is seldom the top performer in DGA tasks due to its sensitivity to feature scaling and noise. In our comparisons, we include k-NN for completeness, with expectations that tree-based or kernel methods will likely surpass it.

2.5.5 Logistic Regression (LR).

Definition: LR is a linear statistical model that predicts class probabilities using a logistic function. It can be extended to multi-class problems via one-vs-rest or softmax (Reference details not provided, n.d.-j).

DGA Applications: Logistic regression has been less commonly used in recent DGA ML studies (due to its linear nature), but it provides a simple probabilistic output. It may serve as a baseline or for binary healthy/fault classification (Reference details not provided, n.d.-f).

Strengths: LR is fast to train and outputs easily interpretable coefficients (each gas feature's weight). It is robust to overfitting if regularized (L1/L2 penalty). Class probabilities are directly obtained, which is useful when comparing classes of unequal importance (Reference details not provided, n.d.-g).

Limitations: Being a linear model, LR cannot capture nonlinear interactions between gases (which are known to exist in faults). Its accuracy is generally lower on complex DGA data compared to nonlinear models. It also assumes independence between features given the log-odds, which may not hold. For multi-class DGA problems (≥ 3 fault

types), one-vs-rest or multinomial logistic regression can be applied, but this increases complexity (Reference details not provided, n.d.-m).

Linking: Logistic Regression is included mainly for benchmarking; one expects non-linear classifiers (SVM, RF, boosting) to outperform it on multi-fault classification tasks due to the complex gas patterns. Nevertheless, its simplicity and probability output can be useful in some ensemble schemes (Reference details not provided, n.d.-f, n.d.-j).

2.5.5 2.5.6 Gradient Boosting Machines (GBM)

Definition: GBM refers to algorithms that build an ensemble of decision trees sequentially, where each new tree tries to correct the errors of the previous ensemble. XGBoost is a popular GBM implementation.

DGA Applications: Gradient boosting (especially XGBoost) is increasingly popular in DGA research for its high predictive power. It effectively learns small residuals, handling heterogeneous data well. Several recent works on transformer DGA classification use XGBoost or LightGBM, sometimes as the final model after data preprocessing or feature selection.

Strengths: Boosting achieves very high accuracy by combining many weak trees. It often wins ML competitions and can handle missing values and mixed types. Feature importances and SHAP values are readily computed for boosted trees.

Limitations: GBM models can be prone to overfitting if not carefully regularized (depth, learning rate). Training can be slower than RF, although implementations like XGBoost and LightGBM optimize speed. Like RF, the final ensemble is not easily interpretable, which motivates the use of SHAP for explanation.

Linking: GBM methods strike a good balance between accuracy and (post-hoc) interpretability. We will include a generic GBM and specifically XGBoost (below) in our comparisons.

2.5.6 2.5.7 XGBoost

Definition: XGBoost is a specific gradient boosting implementation optimized for speed and performance. It builds trees to optimize a user-defined loss function, often with regularization.

DGA Applications: XGBoost has been successfully applied to DGA fault classification.

432 For example, Udeh *et al.* (2026) benchmarked nine models and found that XGBoost
433 achieved the highest macro-F1 (≈ 0.757) on a multi-class DGA task, outperforming SVM
434 and even RF. Its built-in feature importance and compatibility with SHAP make it at-
435 tractive for transformer data.

436 **Strengths:** As a high-capacity learner, XGBoost often yields the best classification met-
437 rics in transformer fault studies. It includes many tuning options (subsampling, column
438 sampling, tree depth) to mitigate overfitting. Training is fast on modern hardware.

439 **Limitations:** XGBoost models are large and opaque without explainability tools. They
440 require parameter tuning (learning rate, tree count, max depth). On very small datasets,
441 they may overfit. Moreover, XGBoost’s strength on high-level fault classes may come at
442 the cost of struggling to discriminate between subtle “moderate” fault levels, as shown
443 by misclassification results in Udeh *et al.*.

444 **Linking:** Given its strong performance, XGBoost is expected to be among the best clas-
445 sifiers for DGA faults. Our third objective will focus on interpreting the top-performing
446 model (likely XGBoost or RF) using SHAP values to identify which gas features drove its
447 decisions.

448 2.6 2.6 Data Challenges and Evaluation Metrics

449 Transformer DGA datasets pose several challenges. **Dataset size and class imbal-**
450 **ance:** Real-world DGA data are often limited (hundreds to a few thousand records)
451 and highly skewed: common faults (or normals) far outnumber rare incipient faults.
452 For instance, Udeh *et al.* used 1,097 records and found over 80% of “moderate fault”
453 samples misclassified as normal, highlighting a “moderate-class bottleneck.” Class im-
454 balance can inflate accuracy if not addressed (e.g. via stratified sampling or weighted
455 loss). **Missing or noisy values:** Field gas measurements may have missing entries
456 or detection limits. Common practice is to impute missing gases (e.g. k-NN imputa-
457 tion) or exclude severely incomplete samples. **Label consistency:** Transformer fault
458 labels (codes for thermal (T1–T3), partial discharge (D1–D2), etc.) may vary by data
459 source or interpretation method. Ensuring consistent labeling (merging similar classes,
460 removing ambiguous cases) is crucial before training. These data issues must be man-
461 aged carefully (e.g. using data resampling techniques or robust training) to avoid biased
462 models.

463 **Evaluation metrics:** To evaluate classifiers, standard metrics include **accuracy** (over-

all correct fraction), **precision** and **recall** for each class, and the **F1-score** (harmonic mean of precision and recall). In multi-class settings with imbalance, *macro-F1* (average F1 across classes) is recommended to treat each class equally. The **confusion matrix** provides class-wise error breakdown. For probabilistic classifiers, **ROC curves** and **area under the ROC curve (AUC)** measure discrimination for each class (often averaged in one-vs-rest fashion). Many DGA studies report accuracy and AUC; for example, Ekojono *et al.* used accuracy, AUC, precision, recall and F1 in a 10-fold CV. In line with these, we will use accuracy, precision, recall, F1, macro-F1, confusion matrices, and class-wise AUC as our evaluation suite.

Table 2.2. DGA feature types.

Feature			
Type	Example Features	Usage	Reference Examples
Gas Concentrations	H ₂ , CH ₄ , C ₂ H ₆ , C ₂ H ₄ , C ₂ H ₂ , CO, CO ₂ (ppm)	Direct indicators of fault gases; used as raw input	Mirowski & LeCun (2012) ; Suwarno et al.
Gas Ratios	CH ₄ /H ₂ , C ₂ H ₆ /CH ₄ , C ₂ H ₄ /C ₂ H ₆ , C ₂ H ₂ /C ₂ H ₄	Capture relative gas patterns (e.g. Rogers ratios)	Rogers (1978) method; Doernenburg method
Combined/ Derived	Log(H ₂), TDCG (sum), gas% of TDCG, normalized features	Enable ML models to learn compound relations; reduce skew	Mirowski & LeCun (2012) (log-transform)
Transformer Metadata	Age, Load, Rated Power	Controls for transformer-specific factors	Abdel-Fattah & Allam (2021)* (example usage of age)
Expanded Gas Ratios	C ₂ H ₂ /H ₂ , CO ₂ /CO, O ₂ /N ₂	Additional ratios to detect leakage or cellulose aging	H2Scan (2024) guide [†]

In some studies, transformer operating data are included to contextualize DGA.

[†]Tutorial example (not peer-reviewed) showing CO₂/CO, O₂/N₂ usage.

476 2.7 2.7 Explainable AI (XAI) for DGA Fault Models

477 Modern ML models achieve high accuracy but are often “black boxes.” Explainable AI
478 (XAI) techniques aim to attribute model decisions to input features. The most prominent
479 XAI tool for DGA models is **SHAP (SHapley Additive exPlanations)**. Lundberg and
480 Lee (2017) introduced SHAP as a unified additive explanation framework: “SHAP assigns
481 each feature an importance value for a particular prediction”. For tree-based models
482 (RF, XGBoost), **TreeSHAP** provides fast, exact computation of SHAP values. The result
483 is a per-sample breakdown of how each gas feature contributed to the predicted class
484 log-odds, grounded in Shapley values from cooperative game theory. SHAP has the
485 desirable properties of consistency (if a model relies more on a feature, its attribution
486 is higher) and local accuracy.

487 **XAI in Transformer DGA Studies:** Recently, SHAP has been applied to interpret
488 DGA models. For example, Suwarno *et al.* (2024) used TreeSHAP on a Random Forest
489 DGA classifier; their analysis showed that feature importances aligned with known gas–
490 fault physics (e.g. high C_2H_2 driving arcing predictions). Similarly, Udeh *et al.* (2026)
491 analyzed their XGBoost DGA model with SHAP and found that the model’s behavior
492 mirrored physical expectations, but also revealed a “gray zone”: most moderate-severity
493 faults had gas levels too close to normal for confident classification. In that study, 83.3%
494 of moderate faults were predicted as normal, and SHAP confirmed that these cases lay
495 in an overlapping gas-space where the model had “limited discriminatory power”. These
496 examples illustrate XAI’s ability to validate model reasoning against domain knowledge
497 and to pinpoint weaknesses (e.g. moderate faults).

498 **Table 2.4. XAI studies on transformer DGA models.**

Study	Model	XAI Method	Key Insights
Udeh <i>et al.</i> (2026)	XGBoost (tree ensemble)	TreeSHAP (SHAP)	Found that model's feature attributions matched physics (e.g. high CH ₄ /C ₂ H ₂ for arcing), but most moderate (incipient) faults had overlapping gas features; SHAP highlighted this "diagnostic gray zone".
Sutikno <i>et al.</i> (2024)	Random Forest	TreeSHAP (SHAP)	(Reported) Feature importances agreed with expected gas–fault links (e.g. H ₂ important for PD), confirming model reliability. (SHAP values revealed which gases drove each fault prediction; details in Heliyon paper).

Note: The above transformer-specific XAI studies demonstrate how SHAP can explain ML fault models and reveal data-driven limitations. Lundberg & Lee (2017) provides the theoretical SHAP framework, and Lundberg *et al.* (2019) extends it with fast tree algorithms.

2.8 Research Gap and Chapter Summary

The reviewed literature shows that many DGA-ML studies have applied standard algorithms and reported accuracy improvements. However, several gaps remain. First, **feature optimization** is often ad hoc: few works systematically evaluate which gas/ratio subset best suits each algorithm. Second, **comparative benchmarks** are lacking: Mirowski & LeCun (2012) and Ekojono *et al.* (2022) lamented the absence of studies that test all key classifiers on a common DGA dataset. Third, while some studies report SHAP analyses, a dedicated application of XAI to pinpoint *which* features are most influential in an ML-based transformer diagnosis model is rare. In particular, no study to our knowledge has coupled a rigorous cross-algorithm comparison (including SVM, DT, RF, KNN, LR, GBM, XGB) with SHAP explanations of the best model's decisions. This identifies our research gap: we must find the optimal gas/ratio feature set for each classifier, compare these algorithms' fault-classification performance, and use SHAP to interpret the top model's feature dependencies. These aims directly address Objectives

518 (1)–(3).

519 **Table 2.3. Performance of ML algorithms on transformer DGA fault data (se-**
520 **lected studies).**

Algorithm	Performance (accuracy/F1/AUC)	Reference
SVM	~78–80% accuracy (linear SVM in one study)	Ekojono <i>et al.</i> (2022)
Decision Tree	~79–80% accuracy (simple DT baseline)	Ekojono <i>et al.</i> (2022)
Random Forest (RF)	~85–90% accuracy (best performer)	Ekojono <i>et al.</i> (2022)
k-Nearest Neighbor	~85.7% (with k=?)	Ekojono <i>et al.</i> (2022)
Logistic Regression	(typically lower; e.g. ~70–75%)	[not directly reported]
Gradient Boosting	–	–
XGBoost	Macro-F1 ≈ 0.757 ; severe-fault AUC ≥ 0.88	Udeh <i>et al.</i> (2026)

521 *Notes:* Performance values vary by dataset and preprocessing. The table highlights
522 that RF and XGBoost consistently rank at the top of reported metrics. A comprehensive
523 comparison on a shared dataset is needed, motivating our structured evaluation.

524 **Appendix A: Dataset**

525 This appendix provides detailed information about the dataset used in the study. It
526 includes descriptions of the ethical approvals obtained, the participants involved, and
527 the experimental design employed.

528 **Ethics Approval**

529 Before commencing the study, ethical approval was obtained from the Institutional Re-
530 view Board (IRB) at [University Name]. The approval ensured that the research adhered
531 to ethical guidelines, protecting the rights and welfare of all participants. The IRB ap-
532 proval number is [Approval Number], and the date of approval is [Approval Date]. All
533 participants provided informed consent before participating in the study.

534 **Participants**

535 The study involved a total of [Number] participants, recruited from [Source/Location].
536 The participants were selected based on specific inclusion and exclusion criteria to en-
537 sure a representative sample. The inclusion criteria included [list criteria], while the
538 exclusion criteria included [list criteria]. The demographic breakdown of the partici-
539 pants is as follows: [Detail demographic information such as age, gender, education
540 level, etc.].

541 **Experimental Design**

542 The experimental design was structured to investigate [briefly describe the objective].
543 The study utilized a [describe the type of design, e.g., randomized controlled trial,
544 between-subjects design, etc.]. Participants were randomly assigned to [number]
545 groups: [describe the groups]. Each group underwent [describe the procedure or
546 intervention]. Data were collected using [describe data collection methods, e.g.,
547 surveys, observations, measurements] at multiple time points: [list the time points].
548 The primary variables measured included [list primary variables]. The design ensured
549 that the study was robust and capable of addressing the research questions effectively.

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581

APPENDIX A

582

SUPPLEMENTARY MATERIALS

583

This is the first appendix.